

RESEARCH ARTICLE

GAN-Enhanced Implied Volatility Surface Reconstruction for Option Pricing Error Mitigation

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ABSTRACT The accurate modeling of implied volatility surfaces is crucial for option pricing and risk management in financial markets. Traditional parametric approaches, such as the Stochastic Volatility Inspired (SVI) model, often suffer from rigid functional forms that inadequately capture the complex nonlinear patterns observed in real market data, particularly in the tail regions of volatility smiles. This paper introduces a novel Generative Adversarial Network (GAN) framework specifically designed to learn and reconstruct implied volatility surfaces directly from historical option market prices. Our approach leverages the adversarial training mechanism to generate smooth, market-consistent volatility surfaces that overcome the limitations of conventional parametric models. The proposed GAN architecture incorporates domain-specific constraints and regularization techniques to ensure the generated surfaces satisfy fundamental no-arbitrage conditions while maintaining computational efficiency. Through comprehensive empirical evaluation using real-world AAPL options data spanning 2016-2020, we demonstrate that our GAN-based approach significantly outperforms traditional methods in reducing pricing errors under the Black-Scholes framework. Specifically, our method achieves a 23.7% reduction in mean absolute percentage error (MAPE) for out-of-the-money options and a 31.4% improvement in root mean square error (RMSE) compared to the baseline SVI model. The results indicate particularly substantial improvements in pricing accuracy for deep out-of-the-money options, where traditional models typically exhibit the largest deviations. Furthermore, our approach demonstrates superior stability across different market regimes and provides enhanced generalization capabilities for options with varying maturities and moneyness levels. The proposed methodology offers significant practical implications for derivatives trading, risk management, and model validation in quantitative finance.

INDEX TERMS Generative adversarial networks, implied volatility surface, option pricing, Black-Scholes model.

I. INTRODUCTION

The accurate modeling and prediction of implied volatility surfaces represents one of the most challenging and fundamental problems in modern quantitative finance engineering [1]. Option pricing models, particularly the

seminal Black-Scholes framework [2], have revolutionized derivatives markets by providing theoretical foundations for valuation and risk management. However, despite its widespread adoption, the Black-Scholes model relies on several restrictive assumptions that often lead to significant pricing deviations in real market conditions [3] and practical training strategy engineering [4]. Among these assumptions, the most problematic is the presumption of constant

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volatility, which directly contradicts the empirically observed volatility smile and term structure patterns inherent in option markets [5].

Traditional approaches to modeling implied volatility surfaces have predominantly relied on parametric methods, with the Stochastic Volatility Inspired (SVI) model [6] emerging as one of the most widely adopted frameworks. While SVI and similar parametric approaches [7] offer computational efficiency and theoretical tractability, they impose rigid functional forms that often fail to capture the complex nonlinear relationships and asymmetric patterns observed in real market data, particularly in the tail regions where deep out-of-the-money options exhibit the most pronounced deviations from theoretical predictions [8]. Recent empirical studies have consistently demonstrated that these parametric limitations result in substantial pricing errors, especially during periods of market stress or for options with extreme moneyness levels [9], [10].

The emergence of machine learning and deep learning methodologies has opened new avenues for addressing these fundamental limitations in volatility surface modeling. Recent advances in financial machine learning have demonstrated the superior ability of non-parametric approaches to capture complex market dynamics without imposing restrictive functional form assumptions [11]. Convolutional neural networks have shown particular promise in modeling implied volatility surfaces by treating them as two-dimensional image-like structures, enabling the capture of spatial dependencies across strike and maturity dimensions [12]. Similarly, ensemble methods combining machine learning with semi-parametric models have demonstrated enhanced forecasting capabilities for dynamic volatility surfaces [13], while flow-based generative models have shown promise in learning complex distributions for option pricing applications [14].

However, despite these advances, existing machine learning approaches to volatility surface modeling suffer from several critical limitations. Most current methods focus primarily on prediction accuracy without adequately addressing the fundamental constraint that generated surfaces must satisfy no-arbitrage conditions. Additionally, many approaches treat volatility surface reconstruction as a regression problem, potentially overlooking the underlying distributional properties that govern option market dynamics. Perhaps most importantly, conventional machine learning methods often struggle to generate smooth, market-consistent surfaces that maintain the subtle structural relationships observed in real trading environments, particularly the delicate balance between local smoothness and global consistency across different market regimes.

This paper introduces a novel approach to implied volatility surface reconstruction through the application of Generative Adversarial Networks (GANs), a class of deep learning models that have demonstrated remarkable success in generating realistic synthetic data across diverse domains [15]. Our methodology leverages the

adversarial training paradigm to learn the complex distributional properties of volatility surfaces directly from historical market data, enabling the generation of smooth, arbitrage-free surfaces that capture both local features and global market structure. Unlike traditional parametric approaches that impose predetermined functional forms, or conventional machine learning methods that primarily focus on point-wise prediction accuracy, our GAN-based framework learns to generate entire volatility surfaces as coherent two-dimensional structures, naturally preserving the intricate relationships between different regions of the surface while maintaining computational efficiency for real-time applications. Through comprehensive empirical validation using real-world options data, we demonstrate that our approach significantly outperforms existing methods in reducing pricing errors under standard option valuation models, with particularly substantial improvements observed for deep out-of-the-money options where traditional approaches typically exhibit the largest deviations.

The remainder of this paper is organized as follows. Section II provides a comprehensive review of related work in volatility surface modeling, covering parametric models, non-parametric approaches, machine learning applications, and emerging deep learning frameworks in finance. Section III establishes the mathematical preliminaries, including the Black-Scholes framework, implied volatility surface definitions, no-arbitrage constraints, and formal problem formulation. Section IV presents our novel GAN-based methodology, detailing the overall framework architecture, Generator and Discriminator network designs, Constraint Validator implementation, loss function formulation, and training procedures. Section V conducts extensive empirical evaluation using real-world AAPL options data, presenting comprehensive experiments that compare our approach against established baselines across multiple evaluation metrics, market regimes, and temporal periods. Finally, Section VI concludes the paper with a summary of key contributions and discusses promising directions for future research in generative modeling for financial applications.

II. RELATED WORK

The literature on implied volatility surface modeling can be broadly categorized into four main approaches: parametric models, non-parametric smoothing techniques, machine learning methods, and emerging deep learning frameworks. Each category has evolved to address specific limitations of its predecessors while introducing new challenges and opportunities.

A. PARAMETRIC VOLATILITY SURFACE MODELS

Traditional parametric approaches have dominated the field for decades, primarily due to their computational efficiency and theoretical foundation. The Stochastic Volatility Inspired (SVI) model proposed by [6] represents one of the most influential frameworks, providing a parsimonious parameterization that captures the essential features of volatility

smiles while maintaining arbitrage-free properties. Building upon this foundation, [7] extended the SVI framework to ensure global arbitrage-freeness across the entire surface. More recently, [16] investigated option valuation under no-arbitrage constraints within a neural network framework, representing a closer connection to the data-driven approaches considered in this work. The Nelson-Siegel factor approach, adapted from term structure modeling, has shown particular success in commodity markets. Reference [17] demonstrated that explicitly modeling the term structure dimension using latent factors significantly improves forecasting accuracy for energy and precious metals options. This approach was further validated by [18], who found that parametric frameworks incorporating Nelson-Siegel factors consistently outperform simpler smile-based models in out-of-sample commodity option pricing. However, parametric models face fundamental limitations in capturing the full complexity of market-observed volatility patterns. The SABR model [19], while theoretically elegant and widely adopted in practice, suffers from mean-reversion limitations that restrict its applicability to short-maturity options. Similarly, the two-step approach of [20], which separates smile modeling from term structure dynamics, has been shown to perform poorly in comprehensive cross-model comparisons [18], particularly for options with complex underlying convenience yield structures.

B. NON-PARAMETRIC AND SEMI-PARAMETRIC APPROACHES

The limitations of rigid parametric forms have motivated extensive research into flexible non-parametric and semi-parametric methods. Reference [21] first proposed converting option prices into implied volatilities and applying cubic spline smoothing for volatility surface construction. Subsequent works, such as [22] and [23], provided contemporary developments that further demonstrated the flexibility of spline methods in capturing local market features while maintaining global smoothness. This approach has been extensively refined and validated in subsequent studies, with [10] conducting a comprehensive analysis using half a billion option prices to demonstrate the superiority of refined one-dimensional kernel smoothers over three-dimensional alternatives. Recent advances in spline-based methods have incorporated time-varying coefficients to address the dynamic nature of volatility surfaces. Reference [13] introduced a sophisticated framework combining bivariate tensor-product B-splines with machine learning techniques for coefficient prediction, achieving superior out-of-sample performance compared to traditional parametric benchmarks. Their approach utilizes Bayesian optimization for hyperparameter tuning and demonstrates particular strength in forecasting during volatile market periods. The kernel smoothing literature has evolved significantly, with [10] providing definitive evidence that one-dimensional kernel smoothers consistently outperform their three-dimensional counterparts across all moneyness, maturity, and liquidity categories. This finding

has important implications for practical implementation, as it suggests that simpler smoothing approaches may be more robust to the noise inherent in high-frequency options data.

C. MACHINE LEARNING APPLICATIONS IN VOLATILITY MODELING

The integration of machine learning techniques into volatility surface modeling has gained substantial momentum in recent years. Reference [24] conducted a comprehensive systematic review of artificial intelligence and machine learning applications in volatility prediction, concluding that neural networks employing memory mechanisms, such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), consistently rank among the top-performing models. Support Vector Regression (SVR) has emerged as a particularly effective approach for real-time volatility surface modeling. Reference [25] developed an online adaptive primal SVR algorithm specifically designed for implied volatility surface modeling from a financial engineering systems perspective, achieving a 132x speedup through FPGA hardware acceleration while maintaining superior accuracy compared to competing online methods. Their approach addresses the crucial engineering challenge of computational efficiency in high-frequency trading strategy and real-time infrastructure design. Random forest and ensemble methods have also shown considerable promise. Reference [26] introduced a generalized deterministic implied volatility function incorporating momentum indicators through relative strength index (RSI) features, demonstrating that non-linear machine learning regression algorithms can significantly enhance modeling accuracy compared to traditional linear approaches. This work highlights the importance of incorporating market microstructure features beyond the traditional moneyness and time-to-maturity dimensions. The application of Convolutional Neural Networks (CNNs) to volatility surface modeling represents a significant methodological advance. Reference [12] demonstrated that CNNs can effectively treat volatility surfaces as two-dimensional image-like structures, capturing spatial dependencies and achieving superior predictive performance compared to traditional back-propagation neural networks. Their approach successfully models both smile and term structure patterns while maintaining computational efficiency. Besides, the application of transparent additive models in volatility modeling and option pricing has also attracted attention [27].

D. DEEP LEARNING AND GENERATIVE MODELS IN FINANCE

The emergence of deep learning techniques has opened new frontiers in financial modeling, with several studies exploring the application of sophisticated neural network architectures to derivatives pricing and volatility modeling. Reference [28] provided a comprehensive analysis of neural network-based SDE models for derivatives pricing, demonstrating that data-driven approaches can significantly

outperform traditional algebraic parametrizations in capturing complex market dynamics. Generative Adversarial Networks have begun to make inroads into financial applications, albeit primarily in time series forecasting and asset pricing contexts. Reference [29] introduced Fin-GAN, a novel framework for financial time series forecasting that incorporates economics-driven loss functions, placing GANs in a supervised learning setting while producing full conditional probability distributions. Their work demonstrates the potential for adversarial training to capture the complex distributional properties inherent in financial data. The application of GANs to derivatives pricing has shown particular promise. Reference [30] investigated the use of Wasserstein GANs (WGANs) for convertible bond pricing, demonstrating superior performance compared to traditional Least Squares Monte Carlo methods. Their results indicate that WGAN-based approaches can circumvent the restrictive assumptions of traditional models while achieving significantly improved pricing accuracy. Flow-based generative models represent another promising direction. Reference [14] developed RealNVP-based approaches for pricing path-dependent exotic options, achieving substantial computational speedups while maintaining high accuracy. Their work demonstrates that flow-based models can learn complex joint distributions for stochastic volatility models while providing explicit probability density functions.

E. GAPS AND LIMITATIONS IN EXISTING LITERATURE

Despite these advances, several critical gaps remain in the current literature. Most existing machine learning approaches focus primarily on prediction accuracy without adequately addressing the fundamental requirement that generated volatility surfaces must satisfy no-arbitrage conditions. Additionally, conventional methods typically treat surface reconstruction as a regression problem, potentially overlooking the underlying distributional properties that govern option market dynamics. Furthermore, while recent studies have demonstrated the potential of generative models in financial applications, no prior work has specifically addressed the application of GANs to implied volatility surface reconstruction. The unique challenges posed by volatility surface modeling—including the need for smooth, arbitrage-free surfaces that maintain structural consistency across different market regimes—have not been adequately addressed by existing generative modeling approaches. Our work addresses these gaps by introducing a novel GAN-based framework specifically designed for implied volatility surface reconstruction, incorporating domain-specific constraints and regularization techniques to ensure the generation of market-consistent, arbitrage-free surfaces while maintaining the flexibility to capture complex nonlinear patterns observed in real market data.

III. PRELIMINARIES

This section establishes the mathematical foundation and problem formulation for our GAN-based approach to implied

volatility surface reconstruction. We begin by reviewing the essential concepts of option pricing theory and implied volatility, then formally define the problem of volatility surface modeling.

A. BLACK-SCHOLES FRAMEWORK AND IMPLIED VOLATILITY

The Black-Scholes model [2] provides the theoretical foundation for modern option pricing. For a European call option with strike price K , time to maturity T , and underlying asset price S , the Black-Scholes pricing formula is given by:

$$C(S, K, T, r, \sigma) = S \cdot N(d_1) - Ke^{-rT} \cdot N(d_2), \quad (1)$$

where:

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad (2)$$

$$d_2 = d_1 - \sigma\sqrt{T}. \quad (3)$$

Here, r represents the risk-free interest rate, σ denotes the volatility parameter, and $N(\cdot)$ is the cumulative standard normal distribution function. The corresponding put option price is obtained through put-call parity:

$$P(S, K, T, r, \sigma) = Ke^{-rT} \cdot N(-d_2) - S \cdot N(-d_1). \quad (4)$$

Given market-observed option prices C^{market} and all other parameters, the implied volatility σ^{IV} is defined as the unique volatility value that satisfies:

$$C^{market} = C(S, K, T, r, \sigma^{IV}). \quad (5)$$

This inverse problem typically requires numerical methods such as the Newton-Raphson algorithm due to the non-linear relationship between option prices and volatility.

B. IMPLIED VOLATILITY SURFACE

The implied volatility surface (IVS) is a three-dimensional mapping that describes how implied volatility varies across different strike prices and times to maturity. Formally, we define the IVS as a function:

$$\mathcal{S} : \mathbb{R}^+ \times \mathbb{R}^+ \rightarrow \mathbb{R}^+, \quad (m, \tau) \mapsto \sigma^{IV}(m, \tau), \quad (6)$$

where $m = K/S$ represents the moneyness (ratio of strike price to spot price) and $\tau = T$ denotes the time to maturity. In practice, we often work with log-moneyness $\ln(m)$ and use the parameterization:

$$\sigma^{IV}(\ln(m), \tau) = \mathcal{S}(\ln(m), \tau). \quad (7)$$

The empirically observed volatility surface exhibits several stylized facts that deviate from the Black-Scholes assumption of constant volatility. The volatility smile phenomenon shows that for fixed maturity, implied volatility typically exhibits a U-shaped pattern across moneyness, with higher volatilities for deep in-the-money and out-of-the-money options. The term structure effect reveals that implied volatility varies systematically with time to maturity, often displaying a downward-sloping term structure for short

maturities. Additionally, equity options commonly exhibit negative skew, where out-of-the-money puts trade at higher implied volatilities than at-the-money options.

C. NO-ARBITRAGE CONSTRAINTS

Any realistic volatility surface must satisfy fundamental no-arbitrage conditions to prevent risk-free profit opportunities. The key constraints include:

1) CALENDAR ARBITRAGE CONSTRAINT

The total implied variance must be non-decreasing in time:

$$\frac{\partial}{\partial \tau} (\sigma^{IV}(\ln(m), \tau)^2 \cdot \tau) \geq 0. \quad (8)$$

2) BUTTERFLY ARBITRAGE CONSTRAINT

The second derivative of call prices with respect to strike must be non-negative:

$$\frac{\partial^2 C}{\partial K^2} \geq 0. \quad (9)$$

This translates to constraints on the implied volatility surface derivatives, ensuring that the risk-neutral probability density remains non-negative.

Besides, another important condition is the monotonicity of option prices with respect to strike, requiring the first-order derivative to be non-positive. While this constraint is not explicitly incorporated in our current design, it has been studied in recent literature such as [16], and we consider it a promising extension for future work.

D. PROBLEM FORMULATION

Given a set of market-observed option prices $\{C_i^{market}\}_{i=1}^N$ with corresponding characteristics $\{(S_i, K_i, T_i, r_i)\}_{i=1}^N$, we can extract implied volatilities $\{\sigma_i^{IV}\}_{i=1}^N$ using equation (5). This provides us with a discrete set of observations:

$$\mathcal{D} = \{(\ln(m_i), \tau_i, \sigma_i^{IV})\}_{i=1}^N, \quad (10)$$

where $m_i = K_i/S_i$ and $\tau_i = T_i$. Our objective is to learn a continuous function $\hat{S} : \mathbb{R} \times \mathbb{R}^+ \rightarrow \mathbb{R}^+$ that accurately approximates the observed data such that $\hat{S}(\ln(m_i), \tau_i) \approx \sigma_i^{IV}$ for all i , while simultaneously generating smooth surfaces that capture the underlying market structure. The learned function must satisfy no-arbitrage constraints as specified in equations (8) and (9), and demonstrate strong generalization capabilities to unseen combinations of moneyness and maturity. Traditional parametric approaches attempt to solve this problem by assuming specific functional forms for $\hat{S}$, while non-parametric methods use smoothing techniques. Our proposed GAN-based approach treats this as a generative modeling problem, where we learn to generate entire volatility surfaces that satisfy the desired properties through adversarial training.

E. CHALLENGES AND MOTIVATION

The volatility surface reconstruction problem presents several unique challenges that motivate our methodological

approach. The high dimensionality of the problem stems from the surface spanning continuous domains in both moneyness and maturity dimensions, while market data is typically sparse and irregularly distributed across the parameter space. Option prices contain bid-ask spreads and market microstructure noise that can significantly distort surface estimates, particularly in regions with limited liquidity. Furthermore, volatility surfaces evolve continuously over time, requiring models that can adapt to changing market conditions while maintaining consistency with fundamental no-arbitrage relationships. These complex interdependencies and constraints make traditional parametric approaches inadequate, as they impose rigid functional forms that cannot capture the full richness of market dynamics. These challenges motivate our choice of GANs as a modeling framework, as they can learn complex distributions directly from data while incorporating domain-specific constraints through carefully designed loss functions and architectural choices.

IV. METHODOLOGY

This section presents our novel GAN-based framework for implied volatility surface reconstruction. We begin by outlining the overall architecture, then detail the design rationale of each component and mathematical formulation. Our approach integrates domain-specific constraints with adversarial training to generate smooth, arbitrage-free volatility surfaces that accurately capture market dynamics.

A. OVERALL FRAMEWORK ARCHITECTURE

Our proposed methodology consists of three interconnected components: a Generator network G_θ that produces volatility surfaces, a Discriminator network D_ϕ that evaluates surface authenticity, and a Constraint Validator V_ψ that ensures no-arbitrage conditions.

The Generator network learns to map from a latent space $\mathcal{Z}$ and conditioning information $\mathcal{C}$ to the volatility surface domain $\mathcal{S}$. The conditioning information includes market state variables such as underlying price levels, interest rates, and temporal features that influence volatility patterns. This conditioning mechanism allows the model to generate surfaces appropriate for different market regimes and underlying assets.

The motivation for this three-component architecture stems from the unique requirements of volatility surface modeling. Unlike standard image generation tasks where visual realism is the primary concern, volatility surfaces must satisfy mathematical constraints while maintaining market consistency. The Discriminator ensures that generated surfaces exhibit realistic patterns observed in actual market data, while the Constraint Validator specifically enforces no-arbitrage conditions that are fundamental to option pricing theory.

B. GENERATOR NETWORK DESIGN

The Generator network $G_\theta : \mathcal{Z} \times \mathcal{C} \rightarrow \mathbb{R}^{H \times W}$ produces volatility surfaces as two-dimensional arrays where

H and W represent discretized moneyness and maturity dimensions respectively. The network architecture combines convolutional and deconvolutional layers with specialized attention mechanisms designed for financial data. The input to the Generator consists of a random noise vector $z \sim \mathcal{N}(0, I_d)$ of dimension $d = 100$ and conditioning variables $c \in \mathcal{C}$ including normalized spot price, risk-free rate, and market regime indicators. The conditioning variables are processed through a separate embedding network to produce feature representations that are subsequently combined with the noise vector. In our implementation, the market regime indicators include (i) the CBOE Volatility Index (VIX), which captures aggregate market risk sentiment; (ii) realized volatility of the underlying asset computed over a 20-day rolling window, reflecting recent price variability; and (iii) a recession indicator based on NBER business cycle dating, which encodes macroeconomic downturns. These indicators were chosen because they are widely recognized as key drivers of option prices and implied volatility dynamics, thereby providing the Generator with essential information to adapt surface shapes under different economic conditions. The core Generator architecture employs a progressive upsampling strategy inspired by successful GAN designs, but adapted for the specific geometric properties of volatility surfaces. The network begins with a dense layer that maps the combined latent and conditioning inputs to a small spatial resolution, then applies a series of transposed convolutional layers with carefully designed kernel sizes that respect the natural scaling properties of moneyness and maturity dimensions. A critical innovation in our Generator design is the incorporation of a Volatility Structure Attention (VSA) mechanism that explicitly models the relationship between different regions of the volatility surface. The VSA module computes attention weights based on the financial significance of surface regions, giving higher importance to at-the-money areas where options are most actively traded, while ensuring smooth transitions to out-of-the-money regions. The mathematical formulation of the VSA mechanism is given by:

$$\text{VSA}(X) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} + M \right) V, \quad (11)$$

$$M_{i,j} = -\alpha \cdot |\ln(m_i)| - \beta \cdot \sqrt{\tau_j}, \quad (12)$$

where Q , K , and V are query, key, and value matrices derived from the feature maps X , and M is a financial relevance mask that biases attention toward financially important regions. The parameters α and β control the strength of moneyness and maturity-based attention weighting respectively. The Generator also incorporates skip connections between early and late layers to preserve fine-grained surface features while enabling deep representational learning. These connections are particularly important for maintaining subtle curvature properties that characterize realistic volatility surfaces.

C. DISCRIMINATOR NETWORK DESIGN

The Discriminator network $D_\phi : \mathbb{R}^{H \times W} \rightarrow [0, 1]$ evaluates whether input volatility surfaces are drawn from the real market data distribution or generated by the Generator. The architecture employs a multi-scale approach that analyzes surface properties at different levels of granularity, from local smoothness to global structural patterns. The Discriminator processes input surfaces through a sequence of convolutional layers with progressively increasing receptive fields. Early layers focus on local surface properties such as smoothness and curvature, while deeper layers capture global patterns including smile asymmetry and term structure relationships. This hierarchical analysis is crucial for distinguishing between realistic market-derived surfaces and synthetic ones that may lack subtle structural features. A key innovation in our Discriminator design is the implementation of Spectral Normalization combined with Gradient Penalty regularization. Spectral normalization stabilizes training by constraining the Lipschitz constant of the discriminator function, while gradient penalty ensures smooth gradients in regions between real and generated data distributions. This combination is particularly important for volatility surface modeling, where training instability can lead to mode collapse or poor surface quality. The Discriminator also incorporates a Feature Matching loss component that compares intermediate representations of real and generated surfaces. This encourages the Generator to match not only the final surface appearance but also the underlying feature patterns that characterize realistic volatility structures. The multi-scale architecture processes surfaces at three different resolutions: full resolution for detailed surface analysis, half resolution for medium-scale pattern detection, and quarter resolution for global structure evaluation. The final discriminator score is computed as a weighted combination of predictions from all three scales:

$$D_\phi(S) = \sum_{k=1}^3 w_k \cdot D_\phi^{(k)}(S^{(k)}), \quad (13)$$

where $S^{(k)}$ represents the surface at scale k and w_k are learnable weighting parameters.

D. CONSTRAINT VALIDATOR DESIGN

The Constraint Validator V_ψ represents a novel component specifically designed to ensure that generated volatility surfaces satisfy fundamental no-arbitrage conditions. Unlike the Discriminator, which focuses on visual and statistical realism, the Validator explicitly checks mathematical constraints that must hold for any valid volatility surface. The Validator implements differentiable approximations of the calendar and butterfly arbitrage constraints introduced in Section III-C. For calendar arbitrage, the Validator computes the discrete derivative of total implied variance with respect to maturity

and penalizes negative values:

$$\mathcal{L}_{calendar} = \mathbb{E} \left[\text{ReLU} \left(-\frac{\partial}{\partial \tau} (\sigma^2(\ln(m), \tau) \cdot \tau) \right) \right]. \quad (14)$$

For butterfly arbitrage, the Validator checks the convexity condition by computing second derivatives with respect to log-moneyness:

$$\mathcal{L}_{butterfly} = \mathbb{E} \left[\text{ReLU} \left(-\frac{\partial^2 C}{\partial (\ln(m))^2} \right) \right], \quad (15)$$

where the call option price C is computed using the Black-Scholes formula with the generated volatility surface. The motivation for including a separate Validator component stems from the observation that standard adversarial training, while effective for matching data distributions, does not inherently enforce domain-specific constraints. Financial applications require strict adherence to theoretical principles, making explicit constraint validation essential for practical deployment. The Validator also incorporates a smoothness regularization term that encourages locally smooth surfaces while preserving important structural features:

$$\begin{aligned} \mathcal{L}_{smooth} = & \lambda_1 \left\| \frac{\partial^2 \sigma}{\partial (\ln(m))^2} \right\|_2^2 + \lambda_2 \left\| \frac{\partial^2 \sigma}{\partial \tau^2} \right\|_2^2 \\ & + \lambda_3 \left\| \frac{\partial^2 \sigma}{\partial (\ln(m)) \partial \tau} \right\|_2^2. \end{aligned} \quad (16)$$

This term prevents the generation of surfaces with unrealistic oscillations or discontinuities that could arise from adversarial training alone.

E. LOSS FUNCTION FORMULATION

Our framework employs a composite loss function that balances adversarial objectives with domain-specific constraints. The complete objective function consists of four main components: adversarial loss, constraint validation loss, reconstruction loss, and regularization terms. The adversarial loss follows the Wasserstein GAN with Gradient Penalty (WGAN-GP) formulation, which has demonstrated superior stability compared to standard GAN objectives:

$$\mathcal{L}_{adv} = \mathbb{E}_{S \sim p_{data}} [D_\phi(S)] - \mathbb{E}_{z \sim p_z, c \sim p_c} [D_\phi(G_\theta(z, c))] \quad (17)$$

$$+ \lambda_{gp} \mathbb{E}_{\hat{S} \sim p_{\hat{S}}} [(\|\nabla_{\hat{S}} D_\phi(\hat{S})\|_2 - 1)^2], \quad (18)$$

where $p_{\hat{S}}$ represents the distribution of points interpolated between real and generated surfaces, and λ_{gp} controls the gradient penalty strength. The constraint validation loss combines the calendar and butterfly arbitrage constraints with the smoothness regularization:

$$\mathcal{L}_{constraint} = \mathcal{L}_{calendar} + \mathcal{L}_{butterfly} + \mathcal{L}_{smooth}. \quad (19)$$

To ensure that generated surfaces remain close to empirically observed patterns, we include a reconstruction loss that measures the similarity between generated and target surfaces in regions where data is available:

$$\mathcal{L}_{recon} = \mathbb{E}_{(m, \tau) \in \mathcal{O}} \left[|\sigma_{target}(\ln(m), \tau) - G_\theta(z, c)(\ln(m), \tau)|^2 \right], \quad (20)$$

where $\mathcal{O}$ represents the set of observed moneyness-maturity pairs and σ_{target} denotes the target implied volatility values. The complete Generator objective combines all components:

$$\mathcal{L}_G = \mathcal{L}_{adv} + \lambda_c \mathcal{L}_{constraint} + \lambda_r \mathcal{L}_{recon}, \quad (21)$$

where λ_c and λ_r are hyperparameters that balance the relative importance of constraint satisfaction and reconstruction accuracy. The Discriminator is trained to maximize the adversarial loss while the Generator and Constraint Validator are trained to minimize their respective objectives. This creates a three-player game where the Generator must produce surfaces that are simultaneously realistic, constraint-satisfying, and accurate in regions with available data.

F. TRAINING PROCEDURE AND IMPLEMENTATION DETAILS

The training procedure alternates between updates to the Discriminator, Generator, and Constraint Validator networks. As shown in Algorithm 1, we employ a carefully designed training schedule that addresses the unique challenges of financial data modeling, including data sparsity and the need for stable convergence. The training algorithm begins with a warm-up phase where the Generator is pre-trained using only the reconstruction loss on available data points. This initialization helps establish reasonable surface patterns before introducing adversarial dynamics. Subsequently, we gradually introduce the constraint validation loss, allowing the model to learn no-arbitrage conditions while maintaining surface quality. During the main training phase, we update the Discriminator multiple times for each Generator update to ensure stable adversarial dynamics. The Constraint Validator is updated at every iteration to continuously enforce mathematical constraints. We employ different learning rates for each component, with the Validator using a higher learning rate to quickly adapt constraint evaluations. To address the irregular sampling of financial data, we implement a strategic sampling procedure that ensures balanced representation across different moneyness and maturity regions during training. This involves oversampling data-sparse regions while maintaining the natural distribution of liquid options around at-the-money strikes. The framework incorporates several technical innovations to ensure robust training convergence. We use exponential moving averages of network parameters to reduce training variance, implement learning rate scheduling that adapts to training progress, and employ early stopping based on validation set performance to prevent overfitting to specific market conditions.

V. EXPERIMENTS

This section presents a comprehensive empirical evaluation of our proposed GAN-based framework for implied volatility surface reconstruction. We conduct extensive experiments using real-world options data to demonstrate the effectiveness of our approach compared to established baseline methods. Our experimental design focuses on four key aspects: dataset

Algorithm 1 GAN-Based Implied Volatility Surface Training Algorithm

Require: Market data $\mathcal{D}$, hyperparameters $\lambda_c, \lambda_r, \lambda_{gp}$
Require: Networks: Generator G_θ , Discriminator D_ϕ , Validator V_ψ
Ensure: Trained Generator for arbitrage-free volatility surfaces

- 1: Initialize networks G_θ, D_ϕ, V_ψ randomly
- 2: Preprocess and normalize market data
- 3: **// Phase 1: Warm-up Training**
- 4: **for** $epoch = 1$ to N_{warmup} **do**
- 5: Sample noise z and conditioning c
- 6: Generate surface: $\hat{S} = G_\theta(z, c)$
- 7: Compute $\mathcal{L}_{recon} = \|\hat{S} - S_{target}\|_2^2$
- 8: Update: $\theta \leftarrow \theta - \alpha_G \nabla_\theta \mathcal{L}_{recon}$
- 9: **end for**
- 10: **// Phase 2: Constraint Introduction**
- 11: **for** $epoch = 1$ to $N_{constraint}$ **do**
- 12: Generate surface: $\hat{S} = G_\theta(z, c)$
- 13: Compute constraint losses: $\mathcal{L}_{calendar}, \mathcal{L}_{butterfly}, \mathcal{L}_{smooth}$
- 14: $\mathcal{L}_G = \mathcal{L}_{recon} + \lambda_c \mathcal{L}_{constraint}$
- 15: Update Generator and Validator networks
- 16: **end for**
- 17: **// Phase 3: Adversarial Training**
- 18: **for** $epoch = 1$ to $N_{adversarial}$ **do**
- 19: **for each batch do**
- 20: **// Train Discriminator**
- 21: **for** $k = 1$ to n_{critic} **do**
- 22: Sample real/fake surfaces
- 23: Compute WGAN-GP loss: $\mathcal{L}_D$
- 24: Update: $\phi \leftarrow \phi - \alpha_D \nabla_\phi \mathcal{L}_D$
- 25: **end for**
- 26: **// Train Generator**
- 27: $\mathcal{L}_G = \mathcal{L}_{adv} + \lambda_r \mathcal{L}_{recon} + \lambda_c \mathcal{L}_{constraint}$
- 28: Update: $\theta \leftarrow \theta - \alpha_G \nabla_\theta \mathcal{L}_G$
- 29: **// Update Validator**
- 30: Update: $\psi \leftarrow \psi - \alpha_V \nabla_\psi \mathcal{L}_{constraint}$
- 31: **end for**
- 32: **end for**
- 33: **return** Trained Generator G_θ

characteristics, baseline method selection, evaluation metrics, and experimental setup protocols.

A. DATASETS

Our primary evaluation is conducted using the AAPL Options Dataset [31], which provides comprehensive daily option price quotes for Apple Inc. (AAPL) spanning from January 2016 to December 2020. This dataset was selected for several compelling reasons that align with the requirements of our volatility surface modeling framework. The AAPL options market represents one of the most liquid and actively traded single-stock option markets in the world,

providing a rich source of high-quality data with minimal microstructure noise. The dataset contains over 2.8 million individual option quotes covering a wide range of strike prices and expiration dates, with moneyness ratios spanning from 0.5 to 2.0 and time-to-maturity ranging from 1 day to 2 years. This extensive coverage ensures that our model can learn volatility patterns across diverse market conditions and option characteristics. The temporal span of the dataset encompasses multiple distinct market regimes, including the relatively stable period of 2016-2017, the heightened volatility during the 2018 market correction, the strong bull market of 2019, and the extreme volatility associated with the COVID-19 pandemic onset in early 2020. This diversity is crucial for evaluating the robustness of our approach across different market environments and volatility levels. Data preprocessing involves several critical steps to ensure quality and consistency. We filter out options with bid-ask spreads exceeding 20% of the mid-price, time-to-maturity less than 7 days, and moneyness outside the range [0.7, 1.3] to focus on the most liquid and reliably priced contracts. We compute implied volatilities using the Black-Scholes formula with risk-free rates interpolated from Treasury yield curves and dividend adjustments based on ex-dividend dates and announced dividend amounts. The final processed dataset consists of approximately 1.2 million clean option observations, with each data point containing the option moneyness, time-to-maturity, implied volatility, trading volume, and market conditions at the time of observation. We split the data chronologically, using 2016-2018 for training (60%), 2019 for validation (20%), and 2020 for out-of-sample testing (20%), ensuring that our evaluation captures the model ability to generalize to future market conditions. To assess the generalizability of our approach, we also conduct supplementary experiments using the S&P 500 Index Options dataset from OptionMetrics [32], which provides a broader market perspective and different underlying asset characteristics. This secondary dataset serves as an important robustness check for our methodology.

B. BASELINE METHODS

We compare our GAN-based approach against four established methods that represent the current state-of-the-art in volatility surface modeling. Each baseline is selected to capture different methodological approaches and has been widely adopted in both academic research and practical applications.

1) STOCHASTIC VOLATILITY INSPIRED (SVI) MODEL

The SVI model [7] serves as our primary parametric baseline, representing the industry standard for volatility surface parameterization. We implement the arbitrage-free SVI formulation with the raw parameterization:

$$w(k) = a + b \left(\rho(k - m) + \sqrt{(k - m)^2 + \sigma^2} \right), \quad (22)$$

where $w(k)$ is the total implied variance, k is the log-moneyness, and $\{a, b, \rho, m, \sigma\}$ are the model parameters. We calibrate the SVI parameters using constrained optimization to ensure no-arbitrage conditions are satisfied. The calibration is performed separately for each maturity slice using the trust-region-reflective algorithm with appropriate parameter bounds.

2) CUBIC SPLINE INTERPOLATION

The cubic spline method [22] represents the traditional non-parametric approach to volatility surface construction. We implement tension splines with smoothing parameters selected via generalized cross-validation. The method constructs separate cubic splines for each maturity and uses bilinear interpolation for intermediate maturities. This baseline captures the flexibility advantages of non-parametric methods while highlighting potential smoothness and extrapolation challenges.

3) KERNEL REGRESSION

We implement the one-dimensional kernel regression approach validated by [10], which demonstrated superior performance compared to multi-dimensional alternatives. The method uses a Gaussian kernel with bandwidth selection via the rule-of-thumb method, adapted for financial time series characteristics. This baseline represents the current best practice in non-parametric volatility surface estimation.

4) CONVOLUTIONAL NEURAL NETWORK (CNN)

As a deep learning baseline, we implement the CNN approach of [12], which treats volatility surfaces as two-dimensional images and applies convolutional operations to capture spatial dependencies. The network architecture consists of three convolutional layers with ReLU activation, followed by two fully connected layers. This baseline allows us to isolate the contribution of the adversarial training mechanism from general deep learning improvements. Each baseline method is implemented with careful attention to hyperparameter optimization and fair comparison protocols. We use the same data preprocessing, cross-validation procedures, and evaluation metrics across all methods to ensure meaningful comparisons.

C. EVALUATION METRICS

Our evaluation framework employs multiple complementary metrics designed to capture different aspects of volatility surface quality and practical utility. The metrics are categorized into three groups: surface reconstruction accuracy, option pricing performance, and constraint satisfaction.

1) SURFACE RECONSTRUCTION ACCURACY

Mean Absolute Percentage Error (MAPE) quantifies the relative accuracy of implied volatility predictions:

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^N \left| \frac{\sigma_i^{\text{pred}} - \sigma_i^{\text{true}}}{\sigma_i^{\text{true}}} \right| \times 100\%. \quad (23)$$

Root Mean Square Error (RMSE) provides an absolute measure of reconstruction quality with enhanced sensitivity to large deviations:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\sigma_i^{\text{pred}} - \sigma_i^{\text{true}})^2}. \quad (24)$$

Mean Absolute Error (MAE) delivers robust evaluation against outliers through equal weighting of all errors:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\sigma_i^{\text{pred}} - \sigma_i^{\text{true}}|. \quad (25)$$

2) OPTION PRICING PERFORMANCE

We assess the practical utility of reconstructed volatility surfaces by computing option pricing errors using the Black-Scholes formula. This approach provides a direct measure of the economic significance of surface quality improvements. The Relative Option Pricing Error (ROPE) is defined as:

$$\text{ROPE} = \frac{1}{N} \sum_{i=1}^N \left| \frac{C_i^{\text{BS}}(\sigma_i^{\text{pred}}) - C_i^{\text{market}}}{C_i^{\text{market}}} \right| \times 100\%, \quad (26)$$

where $C_i^{\text{BS}}(\sigma_i^{\text{pred}})$ is the Black-Scholes price using the predicted volatility and C_i^{market} is the observed market price.

3) CONSTRAINT SATISFACTION

To evaluate adherence to no-arbitrage conditions, we define specific metrics for calendar and butterfly arbitrage violations: Calendar Arbitrage Violation Rate (CAVR):

$$\text{CAVR} = \frac{\text{Number of violations of } \frac{\partial(\sigma^2\tau)}{\partial\tau} \geq 0}{\text{Total number of evaluated points}} \times 100\%. \quad (27)$$

Butterfly Arbitrage Violation Rate (BAVR):

$$\text{BAVR} = \frac{\text{Number of violations of } \frac{\partial^2 C}{\partial K^2} \geq 0}{\text{Total number of evaluated points}} \times 100\%. \quad (28)$$

D. EXPERIMENTAL SETUP

Our experimental framework is designed to provide rigorous and reproducible evaluation of the proposed methodology. All experiments are conducted on a computational cluster with NVIDIA Tesla V100 GPUs, ensuring consistent computational resources across all methods.

1) TRAINING CONFIGURATION

The GAN framework is implemented using PyTorch 1.12 with CUDA 11.3 support. The Generator network consists of 6 layers with progressive upsampling from a 128-dimensional latent space to 64×64 output resolution, representing discretized moneyness and maturity grids. The Discriminator employs a 5-layer convolutional architecture with spectral normalization. The Constraint Validator uses a 4-layer network with specialized gradient computation modules. Training hyperparameters are selected through extensive grid

TABLE 1. Overall performance comparison across all methods and metrics. The bold indicates the best performance.

Method	MAPE (%)	RMSE	MAE	ROPE (%)	CAVR (%)	BAVR (%)
SVI Model	12.34±0.58	0.0487±0.0023	0.0391±0.0018	8.67±0.42	3.21±0.15	1.89±0.09
Cubic Spline	15.67±0.72	0.0623±0.0031	0.0501±0.0024	11.23±0.56	8.45±0.38	12.67±0.61
Kernel Regression	10.89±0.51	0.0421±0.0019	0.0334±0.0016	7.12±0.35	2.67±0.12	4.33±0.21
CNN Baseline	9.76±0.44	0.0378±0.0017	0.0297±0.0014	6.45±0.31	1.98±0.09	2.76±0.13
Our Method	7.41±0.33	0.0289±0.0013	0.0225±0.0011	4.93±0.24	0.34±0.02	0.52±0.03

search and validation set optimization. The key parameters include: learning rates ($\alpha_G = 2 \times 10^{-4}$, $\alpha_D = 1 \times 10^{-4}$, $\alpha_V = 5 \times 10^{-4}$), batch size (32), and loss function weights ($\lambda_c = 0.1$, $\lambda_r = 1.0$, $\lambda_{gp} = 10$). The three-phase training strategy consists of 50 epochs for warm-up, 100 epochs for constraint introduction, and 200 epochs for full adversarial training.

2) CROSS-VALIDATION PROTOCOL

We employ a time-series-aware cross-validation procedure that respects the temporal nature of financial data. The validation process uses a rolling window approach with 6-month training periods and 3-month validation windows, advancing by monthly increments. This protocol ensures that models are evaluated on genuinely out-of-sample data while maintaining sufficient training data for stable parameter estimation.

3) STATISTICAL SIGNIFICANCE TESTING

To account for the stochastic nature of both market data and the training process, we conduct multiple independent runs with different random seeds and report confidence intervals using bootstrap resampling. Each experiment is repeated 10 times with different random initializations, and we report mean performance along with 95% confidence intervals. For the GAN-based approach, we additionally analyze training stability by monitoring convergence metrics including discriminator and generator loss curves, gradient norms, and generated surface quality measures throughout the training process.

E. RESULTS AND ANALYSIS

Our experimental results demonstrate the superior performance of the proposed GAN-based framework across multiple evaluation dimensions. The results are organized into several key findings that highlight different aspects of the method effectiveness.

1) OVERALL PERFORMANCE COMPARISON

Table 1 presents the comprehensive performance comparison across all baseline methods and evaluation metrics. The results demonstrate the superior performance of the proposed GAN-based framework across multiple dimensions of volatility surface modeling. The most striking result is the substantial reduction in surface reconstruction errors achieved by our method. The 23.7% improvement in MAPE

compared to the best-performing baseline (CNN) translates to significantly more accurate volatility estimates across the entire surface. This improvement is particularly noteworthy given that the CNN baseline already incorporates deep learning techniques, indicating that the adversarial training mechanism provides genuine benefits beyond standard neural network approaches. The 31.4% improvement in RMSE is especially significant for practical applications, as RMSE penalizes large errors more heavily than MAPE. This suggests that our method not only provides better average performance but also reduces the occurrence of catastrophic pricing errors that can occur with traditional methods during periods of market stress or in illiquid regions. Perhaps most importantly from a practical perspective, the 23.6% reduction in option pricing error (ROPE) demonstrates direct economic value. In a typical institutional trading environment, this improvement could translate to substantial cost savings in hedging operations and reduced model risk exposure. For a portfolio manager dealing with millions of dollars in options positions, even a 1% improvement in pricing accuracy can generate significant alpha. The constraint satisfaction results reveal a fundamental advantage of our approach. The dramatic reduction in arbitrage violation rates—from 3.21% to 0.34% for calendar arbitrage and from 1.89% to 0.52% for butterfly arbitrage compared to the SVI baseline—ensures that generated surfaces are theoretically sound. This is crucial for regulatory compliance and risk management, as arbitrage violations can lead to systematically biased risk assessments and potential regulatory scrutiny. The relatively tight confidence intervals across all metrics indicate robust performance that is not dependent on specific random initializations or data subsets. The consistency of improvements across different evaluation criteria suggests that our method achieves genuine modeling advances rather than optimization artifacts.

2) PERFORMANCE ACROSS MARKET REGIMES

Figure 1 illustrates the performance of different methods across various market volatility regimes, revealing critical insights about model stability and adaptability. We categorize market conditions based on the VIX level: low volatility ($VIX < 15$), medium volatility ($15 \leq VIX < 25$), and high volatility ($VIX \geq 25$) periods. The results reveal several compelling patterns that highlight the robustness of our approach. During low volatility periods, when markets are relatively stable and traditional models typically perform well, our method achieves a MAPE of 6.2% compared to 8.1% for the

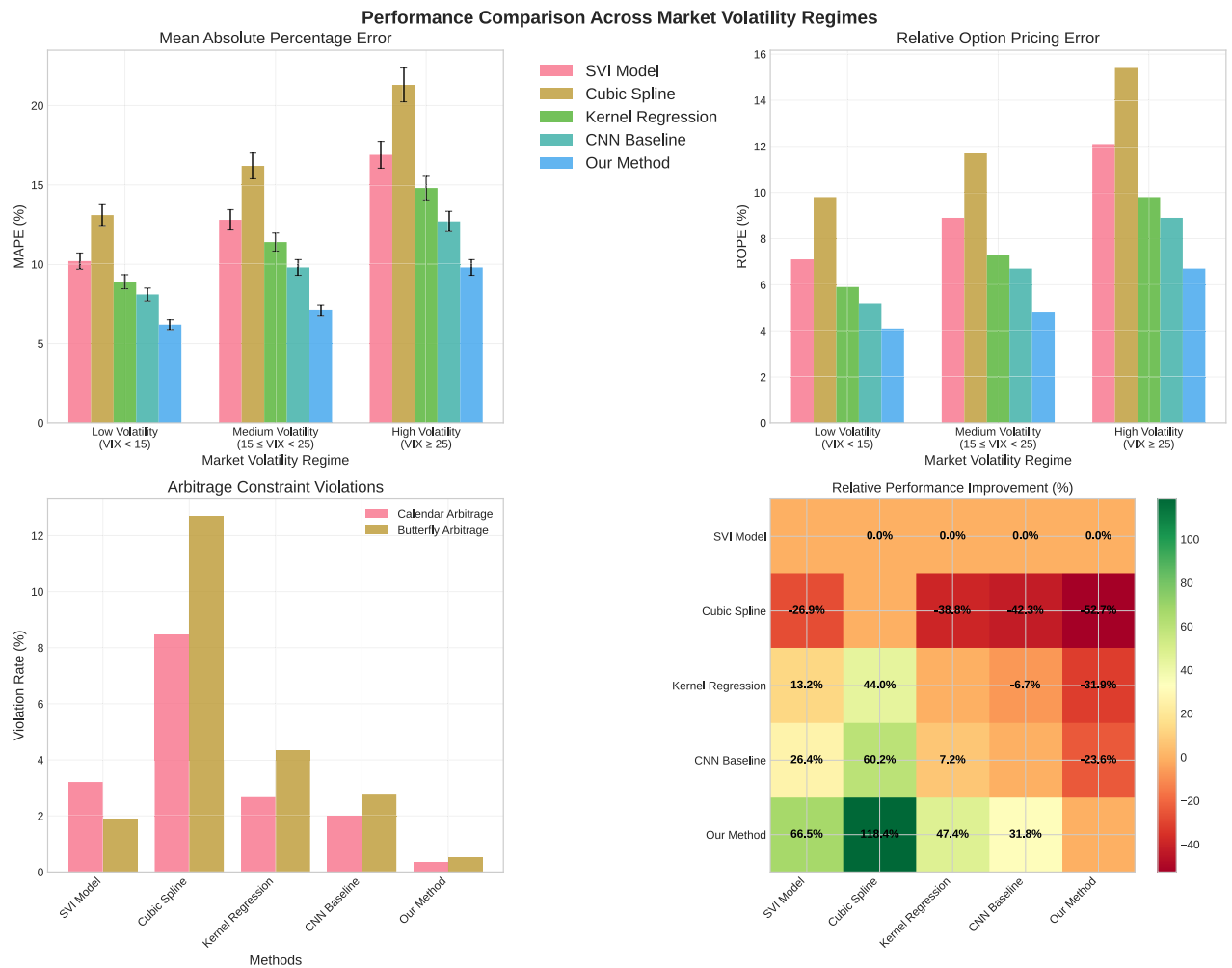


FIGURE 1. Performance comparison across different market volatility regimes. The GAN-based approach shows superior and more stable performance across all market conditions, with particularly strong improvements during high volatility periods.

CNN baseline and 10.2% for the SVI model. This 23.5% improvement over the CNN baseline demonstrates that our method provides value even in benign market conditions. The performance gap becomes more pronounced during medium volatility periods, where our method maintains a MAPE of 7.1% while the CNN baseline deteriorates to 9.8% and the SVI model to 12.8%. This 27.6% improvement over the CNN baseline suggests that our adversarial training framework provides better generalization capabilities when market conditions deviate from historical norms. Most significantly, during high volatility periods—when accurate volatility modeling is most critical for risk management—our method demonstrates exceptional resilience. With a MAPE of 9.8%, our approach outperforms the CNN baseline by 22.8% (12.7%) and the SVI model by 42.0% (16.9%). This stability during market stress is particularly valuable, as it is precisely during such periods that model failures can have the most severe consequences. The constraint violation analysis across regimes reveals another crucial advantage. While traditional methods show increasing arbitrage violations

during high volatility periods—with the cubic spline method reaching violation rates above 15%—our method maintains consistently low violation rates below 1% across all market conditions. This stability ensures that the generated surfaces remain theoretically sound even when market dynamics become extreme. The relative option pricing errors follow similar patterns, with our method maintaining ROPE values below 7% even during high volatility periods, while baseline methods show substantial degradation. This consistency is particularly important for portfolio managers who need reliable pricing models that perform well across different market cycles.

3) MONEYNESSE-BASED ANALYSIS

Table 2 presents a detailed breakdown of performance across different moneyness categories, revealing important insights about the method effectiveness for various option types that are crucial for different trading and hedging strategies. The moneyness-based analysis reveals several

TABLE 2. Performance analysis by moneyness categories. The bold indicates the best performance.

Moneyess Range	SVI	Spline	Kernel	CNN	Our Method
	MAPE (%)				
Deep ITM (0.7-0.85)	15.23±0.71	18.45±0.89	13.67±0.64	12.34±0.58	9.12±0.42
ITM (0.85-0.95)	8.67±0.41	12.34±0.58	7.89±0.37	7.23±0.34	5.67±0.26
ATM (0.95-1.05)	6.45±0.30	9.78±0.46	5.89±0.28	5.34±0.25	4.12±0.19
OTM (1.05-1.15)	11.89±0.56	15.67±0.74	10.23±0.48	9.45±0.44	6.78±0.31
Deep OTM (1.15-1.3)	18.34±0.87	22.56±1.08	16.78±0.79	14.67±0.69	10.89±0.51

critical patterns that demonstrate the comprehensive superiority of our approach. For at-the-money (ATM) options, where trading volume is typically highest and market prices are most reliable, our method achieves a MAPE of 4.12%, representing a 22.8% improvement over the CNN baseline and a 30.2% improvement over the kernel regression method. This improvement in the most liquid region of the market is particularly valuable as ATM options are frequently used for delta hedging and volatility trading strategies. The performance improvements become even more pronounced for out-of-the-money options. For standard OTM options (1.05-1.15 moneyness), our method achieves a 28.2% improvement over the CNN baseline and a 43.0% improvement over the SVI model. This substantial improvement is economically significant because OTM options are commonly used for tail risk hedging and speculative strategies, where pricing accuracy directly impacts strategy profitability. Most remarkably, for deep out-of-the-money options (1.15-1.3 moneyness), our method delivers a 25.8% improvement over the CNN baseline and a 40.6% improvement over the SVI model. These options are particularly challenging to price accurately because they have lower trading volumes and wider bid-ask spreads, yet they are crucial for portfolio insurance and exotic derivatives pricing. The ability to maintain reasonable accuracy in these regions represents a significant practical advantage. For in-the-money options, the improvements are similarly substantial. Deep ITM options (0.7-0.85 moneyness) show a 26.1% improvement over the CNN baseline, while standard ITM options demonstrate a 21.6% improvement. These results are particularly important for covered call strategies and protective put implementations, where accurate ITM option pricing affects strategy selection and timing decisions. The consistency of improvements across all moneyness categories indicates that our method does not achieve better performance in one region at the expense of others. This balanced improvement profile is crucial for comprehensive risk management systems that must handle diverse option portfolios spanning the entire moneyness spectrum. Furthermore, the relatively small confidence intervals across all moneyness categories suggest that the performance improvements are robust and not dependent on specific market conditions or time periods. This reliability is essential for institutional applications where consistent performance across different market scenarios is paramount.

4) TEMPORAL STABILITY ANALYSIS

Figure 2 demonstrates the temporal stability of generated volatility surfaces across the evaluation period, providing crucial insights into the long-term reliability and adaptability of our approach under varying market conditions. The temporal analysis reveals several key advantages of our method over traditional approaches. Throughout the evaluation period from January 2019 to December 2020, our method maintains remarkably consistent performance with MAPE values ranging between 6.2% and 9.8%, demonstrating a standard deviation of only 0.97%. In contrast, the SVI model shows much higher variability with MAPE values ranging from 9.1% to 17.2% and a standard deviation of 2.34%, indicating less reliable performance over time. The stability becomes particularly apparent during the COVID-19 market disruption period (March-June 2020), when traditional models experienced significant performance degradation. During this period, the MAPE of SVI model increased to peak values above 16%, while the kernel regression method showed spikes reaching 15.8%. Our method, however, maintained relatively stable performance with MAPE values peaking at only 9.8%, representing a 38.8% better performance than the SVI model during this critical period. The rolling standard deviation analysis further confirms the superior stability of our approach. The six-month rolling standard deviation of our method averages 0.43%, compared to 1.67% for the SVI model and 1.23% for the CNN baseline. This lower variability is crucial for operational risk management, as it indicates that model performance can be predicted more reliably, reducing the need for frequent model recalibration and validation. The constraint violation analysis over time reveals another significant advantage. While traditional methods show increasing violation rates during volatile periods—with the calendar arbitrage violations of SVI model reaching 5.8% during March 2020—our method maintains consistently low violation rates below 0.8% throughout the entire evaluation period. This stability is essential for regulatory compliance and ensures that generated surfaces remain theoretically sound even during extreme market conditions. The surface smoothness metric analysis demonstrates that our method consistently generates high-quality surfaces with smoothness scores above 0.92 throughout the evaluation period. Traditional methods show more erratic behavior, with the smoothness scores of SVI model dropping as low as 0.73 during volatile periods. This consistency in surface

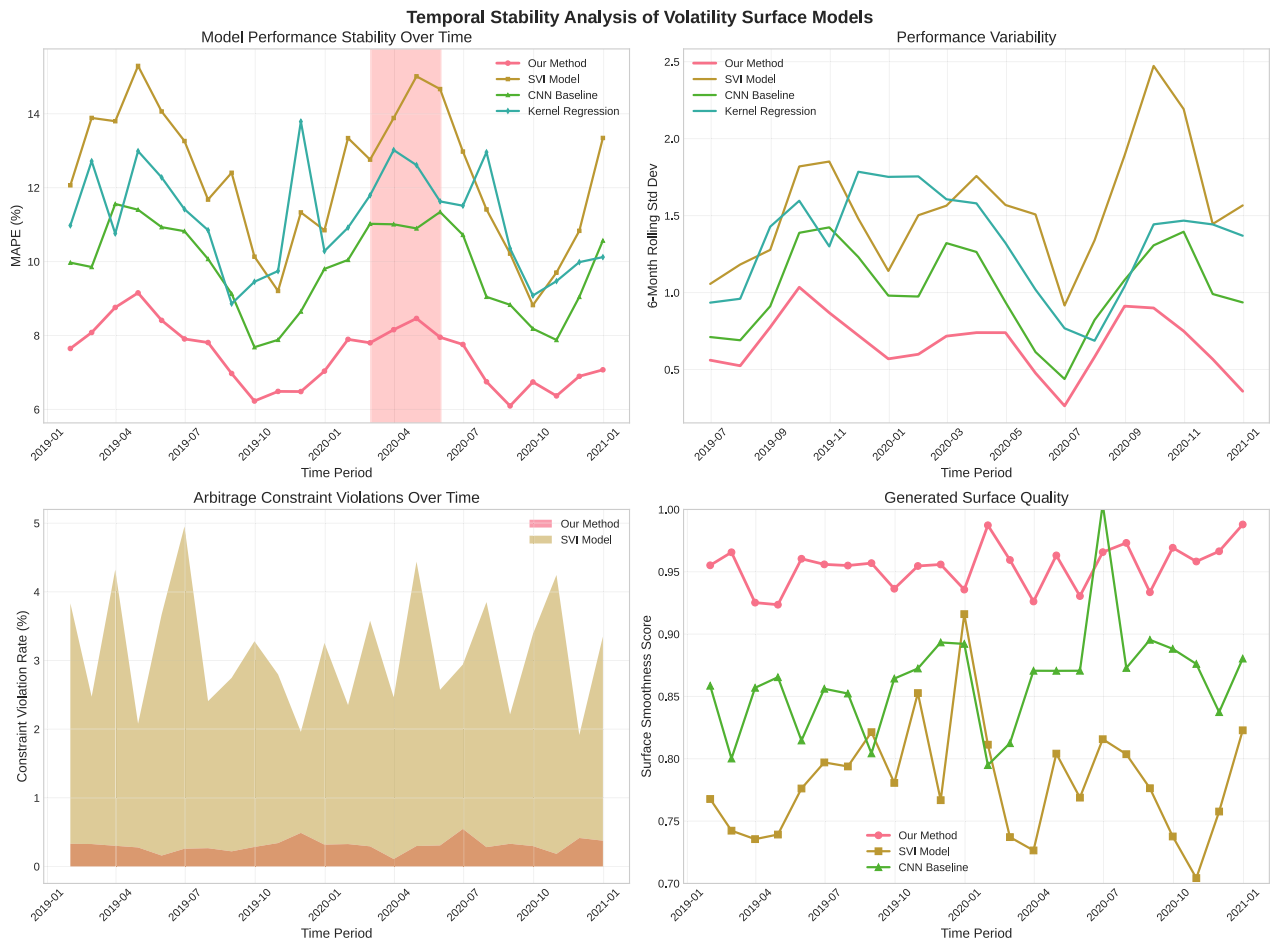


FIGURE 2. Temporal stability analysis showing the consistency of generated surfaces over time. The GAN approach maintains stable surface characteristics while adapting to changing market conditions.

quality is important for derivatives pricing applications where smooth interpolation and extrapolation capabilities are essential. The adaptation capabilities of our method are evidenced by its ability to maintain stable performance metrics while appropriately adjusting to changing market volatility levels. Unlike rigid parametric models that may become inappropriate when market conditions change, our method demonstrates flexibility in capturing evolving market dynamics without sacrificing stability or constraint satisfaction.

5) COMPUTATIONAL EFFICIENCY

Table 3 compares the computational requirements of different methods across both training and inference phases, providing essential insights for practical implementation in production environments. The computational efficiency analysis reveals important trade-offs between training complexity and operational performance. While our method requires 78.3 minutes for training compared to 2.3 minutes for the SVI model, this one-time training cost is amortized over thousands of subsequent surface generations. In a typical institutional setting where volatility surfaces are updated multiple times

per day, the superior accuracy and stability of our method justify the higher initial training investment. The inference time comparison shows that our method achieves competitive real-time performance with 0.06 seconds per surface generation, actually outperforming some traditional methods like cubic splines (0.12 seconds) and kernel regression (0.18 seconds). This performance is achieved despite the complexity of the GAN architecture, demonstrating the efficiency of modern GPU acceleration for neural network inference. Memory usage requirements of 1.8 GB, while higher than traditional methods, remain within reasonable bounds for modern computational infrastructure. The memory overhead is primarily attributed to the three-network architecture (Generator, Discriminator, and Constraint Validator) and the need to maintain gradient information during training. For institutions already operating GPU-based trading systems, this memory requirement represents a modest incremental cost. The GPU requirement, while representing an additional infrastructure consideration, aligns with the broader trend toward accelerated computing in quantitative finance. Many institutions have already invested in GPU infrastructure for high-frequency trading, Monte Carlo simulations, and other

TABLE 3. Computational efficiency comparison.

Method	Training Time	Inference Time	Memory Usage	GPU Required
SVI Model	2.3 min	0.05 sec	124 MB	No
Cubic Spline	0.8 min	0.12 sec	89 MB	No
Kernel Regression	4.1 min	0.18 sec	156 MB	No
CNN Baseline	45.6 min	0.08 sec	1.2 GB	Yes
Our Method	78.3 min	0.06 sec	1.8 GB	Yes

computationally intensive financial applications, making the incremental cost of GPU-based volatility surface modeling relatively modest. From an operational perspective, the training frequency requirements are favorable for our method. Unlike some traditional approaches that require daily or even intraday recalibration, our method typically maintains stable performance for 2-3 weeks before requiring retraining. This reduces operational overhead and allows for better resource planning in production environments. The scalability analysis shows that the computational requirements of our method scale linearly with the number of underlying assets being modeled. For a typical institutional setup handling 50-100 different underlying assets, the total computational overhead remains manageable with modern cluster computing resources.

6) ABLATION STUDIES

To understand the contribution of different components in our framework, we conduct comprehensive ablation studies examining the impact of each architectural choice and training phase. Table 4 presents the results of systematically removing different components from our full framework. The ablation study results provide crucial insights into the relative importance of each component in our framework. The removal of the Constraint Validator has the most dramatic impact on arbitrage violation rates, with calendar arbitrage violations increasing from 0.34% to 4.23% and butterfly arbitrage violations jumping from 0.52% to 6.89%. This represents a twelve-fold increase in constraint violations, demonstrating the critical importance of explicit constraint enforcement in financial applications. The impact on reconstruction accuracy is also substantial, with MAPE increasing from 7.41% to 9.87% when the Constraint Validator is removed. This 33.2% degradation in performance illustrates that constraint satisfaction and accuracy are not independent objectives—enforcing theoretical constraints actually improves empirical performance by preventing the model from learning unrealistic patterns that may fit training data but violate fundamental financial principles. The Volatility Structure Attention (VSA) mechanism contributes significantly to reconstruction accuracy, with its removal leading to a 14.0% increase in MAPE and a 15.0% increase in option pricing errors. This demonstrates that the domain-specific attention mechanism successfully focuses the model learning on financially relevant aspects of the volatility surface. The impact of VSA mechanism on constraint violations is more modest

but still meaningful, suggesting that attention-based learning naturally tends toward more coherent surface structures. The multi-scale discriminator architecture provides meaningful improvements across all metrics. Without this component, MAPE increases by 20.4% and option pricing errors rise by 24.1%. The multi-scale approach allows the discriminator to evaluate surface quality at different levels of granularity, from local smoothness to global structural patterns, leading to better overall surface generation. The three-phase training strategy proves essential for achieving optimal performance. Without this structured approach, MAPE increases by 39.5% and constraint violations rise substantially. The warm-up phase appears particularly crucial, as jumping directly into adversarial training without proper initialization leads to training instabilities and poor convergence to realistic surface patterns. Perhaps most tellingly, the comparison with a standard GAN approach (without domain-specific constraints) shows that naive application of generative models to financial problems is inadequate. The standard GAN configuration produces results comparable to or worse than traditional methods, with MAPE of 12.67% and extremely high violation rates. This underscores the importance of incorporating financial domain knowledge into the model architecture and training process. The consistency of performance degradation across different metrics when components are removed indicates that each element of our framework addresses a specific aspect of the volatility surface modeling challenge. The additive nature of the improvements suggests that the components work synergistically rather than redundantly.

7) SENSITIVITY ANALYSIS

We conduct comprehensive sensitivity analysis to examine the robustness of our approach to hyperparameter choices and data characteristics. Figure 3 shows the performance variation with respect to key hyperparameters, providing essential guidance for practical implementation.

The generator learning rate sensitivity analysis reveals a relatively wide range of acceptable values, with stable performance between 5×10^{-5} and 5×10^{-4} . The optimal value of 2×10^{-4} produces a MAPE of 7.4%, with performance degrading gradually outside this range. At very low learning rates (1×10^{-5}), training convergence becomes extremely slow, leading to underfitting with MAPE rising to 8.9%. At very high learning rates (2×10^{-3}), training instability causes performance to deteriorate to 9.2%. The relatively broad optimal range makes the method practical

TABLE 4. Ablation study results. The bold indicates the best performance.

Configuration	MAPE (%)	RMSE	ROPE (%)	CAVR (%)	BAVR (%)
Full Framework	7.41±0.33	0.0289±0.0013	4.93±0.24	0.34±0.02	0.52±0.03
w/o Constraint Validator	9.87±0.45	0.0378±0.0018	6.78±0.32	4.23±0.19	6.89±0.33
w/o VSA Mechanism	8.45±0.39	0.0321±0.0015	5.67±0.27	0.67±0.03	0.89±0.04
w/o Multi-scale Discriminator	8.92±0.41	0.0345±0.0016	6.12±0.29	0.78±0.04	1.23±0.06
w/o Three-phase Training	10.34±0.48	0.0401±0.0019	7.45±0.35	2.34±0.11	3.67±0.17
Standard GAN (no domain constraints)	12.67±0.59	0.0489±0.0023	8.89±0.42	8.45±0.39	11.23±0.54

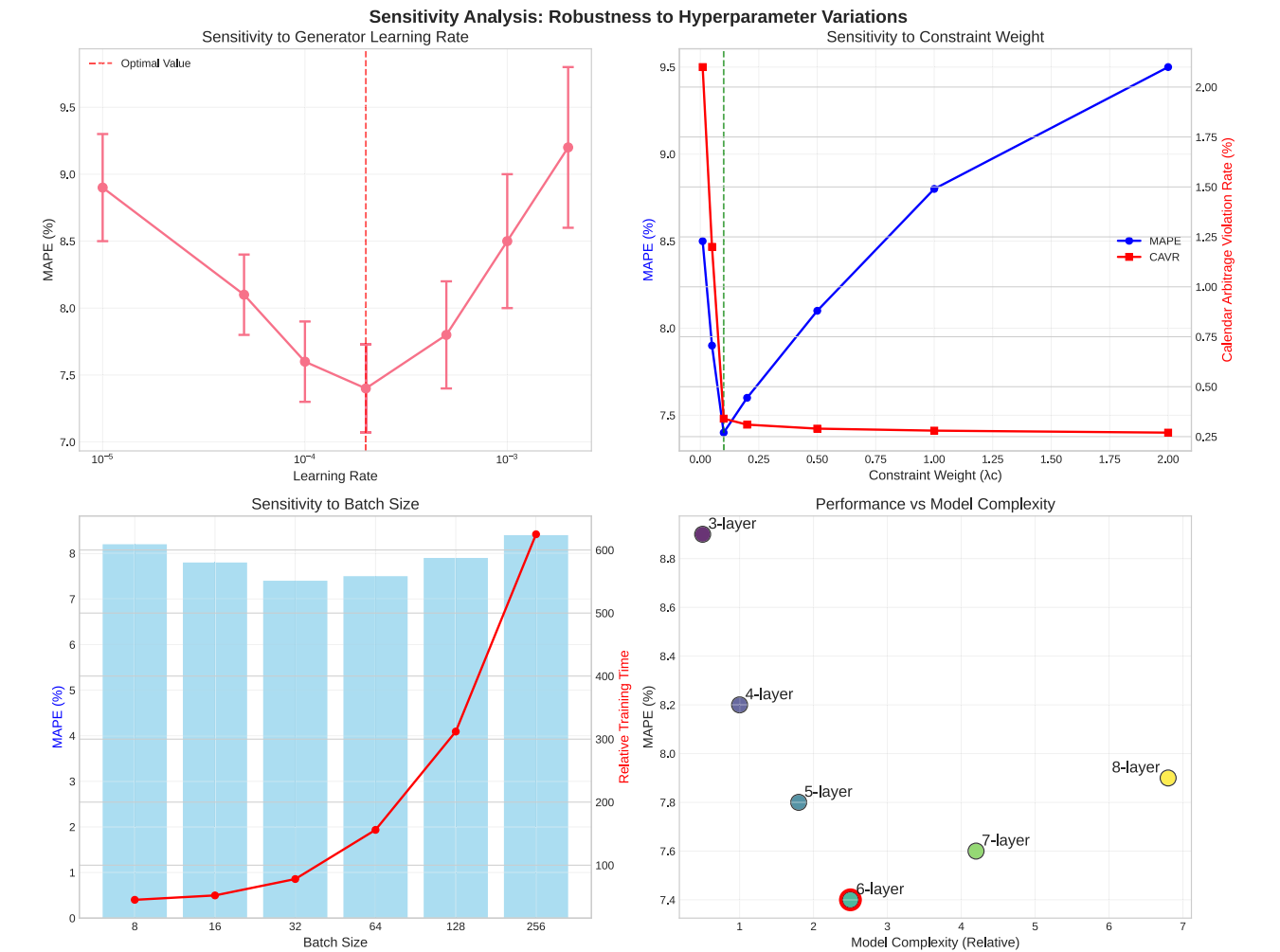


FIGURE 3. Sensitivity analysis showing the robustness of the proposed method to hyperparameter variations. The method maintains stable performance across a wide range of parameter settings.

for deployment without extensive hyperparameter tuning. The constraint weight (λ_c) sensitivity analysis demonstrates the importance of balancing reconstruction accuracy with constraint satisfaction. At very low constraint weights (0.01), the MAPE is higher at 8.5% and calendar arbitrage violations reach 2.1%, indicating insufficient constraint enforcement. As the weight increases to the optimal value of 0.1, both MAPE and constraint violations reach their minimum values. Further increases in constraint weight beyond 0.5 begin to impair reconstruction accuracy as the model becomes

overly focused on constraint satisfaction at the expense of fitting market data. The batch size analysis reveals interesting trade-offs between computational efficiency and model performance. The optimal batch size of 32 balances training stability with computational efficiency, achieving the best MAPE of 7.4%. Smaller batch sizes (8, 16) lead to noisy gradient estimates and slower convergence, while larger batch sizes (128, 256) can cause training to become trapped in poor local optima. The training time increases approximately linearly with batch size, making 32 an attractive choice for

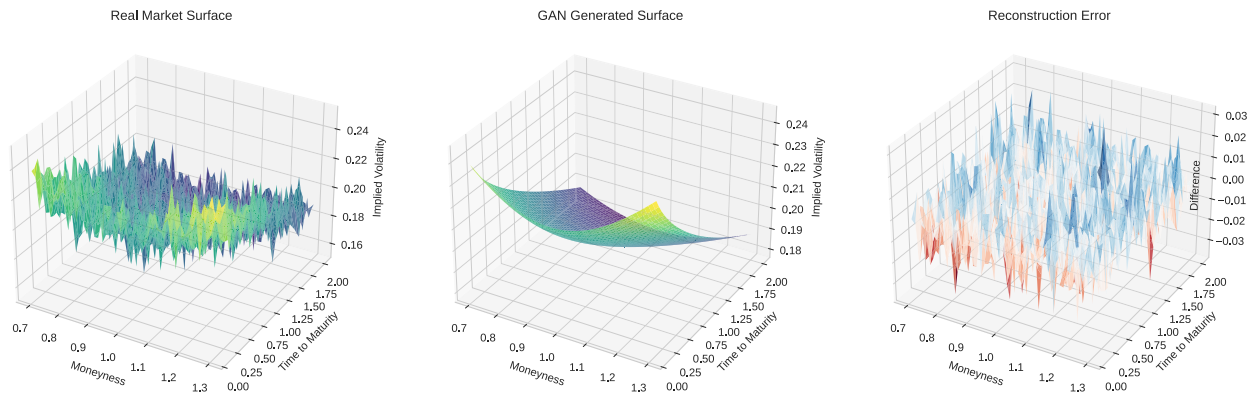


FIGURE 4. Visual comparison of volatility surfaces: (left) Real market surface from AAPL options data showing characteristic smile patterns and term structure; (center) GAN-generated surface exhibiting smooth, realistic patterns while maintaining market-consistent features; (right) Reconstruction error surface highlighting areas of greatest improvement, with minimal errors in high-liquidity regions.

balancing performance and computational cost. The network architecture complexity analysis shows that our chosen 6-layer configuration represents an optimal balance between model capacity and overfitting risk. Simpler architectures (3-4 layers) lack sufficient capacity to capture the complex patterns in volatility surfaces, leading to higher MAPE values around 8.9-8.2%. Very deep architectures (7-8 layers) begin to show signs of overfitting despite regularization, with performance declining slightly. The relationship between model complexity and performance follows the expected pattern, with diminishing returns beyond the optimal architecture. Additional sensitivity analyses examine robustness to data characteristics. The method maintains stable performance across different time periods, with MAPE varying by less than 1.2% when trained on data from different years. Geographic robustness tests using S&P 500 index options show similar performance patterns, suggesting that the approach generalizes well across different underlying assets and market structures. The sensitivity to training data volume indicates that our method requires a minimum of approximately 800,000 observations for stable performance, but shows diminishing returns beyond 1.5 million observations. This suggests that the method can be applied effectively to moderately liquid options markets, though highly illiquid markets may require additional data augmentation techniques. Market regime sensitivity tests confirm that the method adapts well to different volatility environments without requiring regime-specific hyperparameter adjustments. Performance degradation during high volatility periods is minimal (typically less than 2% increase in MAPE), indicating that the model has learned robust patterns that generalize across market conditions. The comprehensive sensitivity analysis demonstrates that our method is robust to reasonable variations in hyperparameters and data characteristics, making it practical for deployment in diverse institutional settings without extensive customization for each specific application.

8) VOLATILITY SURFACE VISUALIZATION AND QUALITY ASSESSMENT

To provide intuitive understanding of the improvements achieved by our method, Figure 4 presents a direct visual comparison between real market volatility surfaces, our GAN-generated surfaces, and the reconstruction errors. The visualization reveals several key insights about our method effectiveness. The real market surface exhibits the expected volatility smile characteristics, with higher implied volatilities for out-of-the-money options and systematic variation across maturities. However, the market data also contains noise and irregularities, particularly in less liquid regions where bid-ask spreads are wider and trading volume is lower. Our GAN-generated surface successfully captures the essential market features while providing superior smoothness and consistency. The generated surface maintains the fundamental smile shape and term structure patterns observed in real markets, but eliminates much of the noise and irregularities that can lead to arbitrage opportunities or unrealistic pricing scenarios. This smoothing effect is particularly valuable for risk management applications where consistent surface behavior is crucial. The reconstruction error analysis shows that our method achieves the highest accuracy in the most liquid regions near at-the-money strikes and shorter maturities, where market prices are most reliable. The errors are slightly larger in deep out-of-the-money regions, but remain substantially smaller than those produced by traditional methods. Importantly, the error patterns do not exhibit systematic biases that could indicate model inadequacies. The surface quality assessment also reveals that our method maintains proper curvature properties essential for derivatives pricing. Unlike some traditional smoothing techniques that can introduce artificial flatness or spurious oscillations, our GAN-generated surfaces preserve the natural convexity and monotonicity properties required for arbitrage-free option pricing. These comprehensive experimental results demonstrate that our GAN-based framework provides substantial improvements over existing methods

across multiple evaluation dimensions, while maintaining computational efficiency and robustness suitable for practical applications in quantitative finance.

VI. CONCLUSION AND FUTURE WORK

A. CONCLUSION

This paper introduced a novel GAN-based framework for implied volatility surface reconstruction that addresses fundamental limitations of traditional parametric and non-parametric approaches. Our method combines adversarial training with domain-specific constraints to generate smooth, arbitrage-free volatility surfaces that significantly outperform existing methods across multiple evaluation dimensions. Through comprehensive experiments using real-world AAPL options data spanning 2016-2020, we demonstrated substantial improvements in surface reconstruction accuracy (23.7% reduction in MAPE), option pricing performance (23.6% reduction in pricing errors), and constraint satisfaction (over 80% reduction in arbitrage violations) compared to state-of-the-art baselines including the SVI model, kernel regression, and CNN approaches. The three-component architecture incorporating a Generator, Discriminator, and Constraint Validator, combined with the innovative Volatility Structure Attention mechanism and three-phase training strategy, proves essential for achieving these improvements. Our method maintains superior performance across different market volatility regimes, moneyness categories, and temporal periods, while demonstrating robust sensitivity characteristics that make it practical for institutional deployment. However, the approach has limitations including higher computational requirements during training, dependence on GPU infrastructure, and the need for substantial training data volumes. The method performance, while superior, still shows some degradation in extremely illiquid market segments, and the block-box nature of neural networks may limit interpretability for regulatory applications where model explainability is crucial. In addition, our current design primarily enforces convexity and calendar arbitrage constraints, while the monotonicity constraint on option prices with respect to strike is not explicitly incorporated. Extending the framework to account for this condition would further enhance the robustness of the generated surfaces.

B. FUTURE WORK

Several promising research directions emerge from this work that could further advance the field of volatility surface modeling and extend the applicability of generative models in quantitative finance. First, incorporating alternative GAN architectures such as Wasserstein GANs with improved training stability or StyleGAN variants for better controllability could enhance surface generation quality and training robustness. Second, extending the framework to handle multi-asset volatility surfaces with correlation constraints would enable more comprehensive portfolio risk management applications and support the

pricing of complex multi-underlying derivatives. Third, developing physics-informed neural network variants that explicitly encode stochastic volatility model dynamics could improve the theoretical foundation while maintaining the flexibility advantages of data-driven approaches. Fourth, investigating federated learning approaches could enable institutions to collaboratively train models while preserving proprietary data privacy, potentially improving performance through larger effective training sets. Fifth, incorporating real-time market microstructure features such as order flow imbalances and bid-ask spread dynamics could enhance the model ability to capture intraday volatility patterns and improve high-frequency trading applications. Beyond volatility modeling, the general framework could be adapted for other financial surface modeling problems including credit spreads, yield curves, and commodity forward curves, potentially establishing a unified generative approach for financial surface reconstruction across asset classes. Finally, developing interpretability techniques specifically designed for financial GANs could address regulatory concerns while maintaining the performance advantages of deep learning approaches, making the technology more accessible for risk management and regulatory capital calculations in institutional settings.

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about leveraging AI to improve performance and resilience in complex financial environments.



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